APPENDIX
I. TEST CASES AND SIMULATION PARAMETERS

A. Test Cases

The large-scale dispatch case uses a balanced AC 69-bus radial feeder with constant-power loads. Line impedances, base loads, and branch current ratings follow the public dataset in [1]. Fig. A1 shows the feeder and microgrid connections: each of nine installation buses hosts 100 PCC–microgrid pairs, yielding 900 PCCs and 2,700 resource units. The model contains 10,800 continuous controls and 900 binary reactive-support modes. PCC exchanges are coupled through feeder voltages, branch currents, and losses.

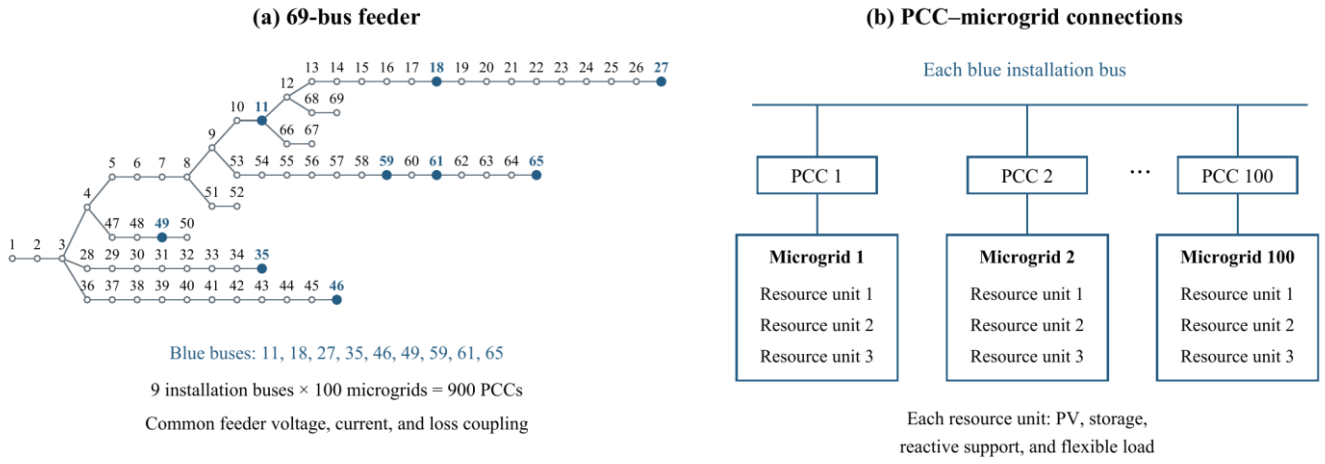


Fig. A1. The 69-bus, 900-PCC test system and the microgrid connections at each installation bus.

B. Simulation Parameters

The network uses 12.66 kV and 10 MVA bases, with bus 1 fixed at 1.00∠0° p.u. Aggregate resource ratings are allocated among the microgrids at each installation bus. Table A1 lists the feeder and resource settings. Operating profiles use DE_KN_residential3 in the OPSD Household Data package [2].

TABLE A1 FEEDER AND RESOURCE PARAMETERS

Parameter	69-bus, 900-PCC configuration
Radial branches / installation buses	68 / 9
Base active / reactive feeder load	3.80189 MW / 2.69410 Mvar
Base branch current limits	200 / 300 / 400 A, as specified by branch [1]
Aggregate PV active power rating	8.0 MW
Aggregate storage power / energy rating	2.8 MW / 7.0 MWh
Aggregate reactive support / load-shedding rating	3.4 Mvar / 0.8 MW
Additional aggregate microgrid load base	1.2 MW
Voltage limits / dispatch interval	0.95–1.05 p.u. / 0.25 h
Storage SOC limits; charge / discharge efficiency	0.20–0.90; 0.95 / 0.95
PV inverter apparent power rating	1.10 × rated PV active power
Base exchange bounds per PCC	[−13, 10] kW active; [−8, 8] kvar reactive

The LLM compiles operating requirements into typed Rules and coordinates Skills through three parallel strategy subagents. The service is deepseek-flash (temperature 0.2, thinking disabled); rule and strategy output limits are 1200 and 600 tokens. Numerical execution uses one thread and a 90 s budget. Sparse elastic SQP allows 50 iterations; Clarabel allows 150 iterations with tolerance 10^{-9}. Independent AC power-balance verification uses a 10^{-8} p.u. tolerance. Device, SOC, and independently checked task constraints use 10^{-8} in their respective scales; normalized network inequalities use 10^{-7}.

Initial SOC is sampled in 0.42–0.75 (0.205 for restricted units). Base cost weights are 0.8–1.2 for storage, 0.04 for reactive power, 5–7 for load shedding, and 4 for PV curtailment; the reactive-support activation cost is 0.004 per PCC group. Positive PCC active power denotes net import.

DATA SOURCES

[1] xiaoniumang/pandapower, "ieee69bus.txt," 69-bus line and load data. [Online]. Available: https://github.com/xiaoniumang/pandapower/blob/e6e834b2d0312da4994c01d24d5605508bc04fd2/ieee69bus.txt

[2] Open Power System Data, "Household Data," Apr. 15, 2020, DE_KN_residential3. [Online]. Available: https://data.open-power-system-data.org/household_data/2020-04-15/